Linear Graphs 1: Tables and the Equation

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KEY WORDS table of values *(tūtohi uara)* · gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c

A Reading a table ●○○

Remember: In a table for a straight line, y changes by the same amount each time x goes up by 1. That amount is the gradient m. The value of y when $x = 0$ is the y-intercept c.

1. The table shows some of the points on a straight line.

x	-2	-1	0	1	2	3
y	-4	-1	2	5	8	11

a) What is y when $x = 2$? _____

b) Each time x goes up by 1, what happens to y? _____

c) What is y when $x = 0$? _____

d) Continue the pattern: what is y when $x = 4$? _____

e) Write the equation of the line. _____

2. The table shows some of the points on a straight line.

x	-1	0	1	2	3	4
y	7	5	3	1	-1	-3

a) What is y when $x = 3$? _____

b) Each time x goes up by 1, what happens to y? _____

c) What is y when $x = 0$? _____

d) Continue the pattern: what is y when $x = 5$? _____

e) Write the equation of the line. _____

B A table from the equation ●○○

Remember: Put each value of x into the equation to find y. Take care with negatives: $2 \times (-3) = -6$.

3. Complete the table of values for each line.

a) $y = 2x - 3$

x	-2	-1	0	1	2
y					

b) $y = -x + 4$

x	-2	-1	0	1	2
y					

c) $y = 4 - 3x$

x	-2	-1	0	1	2
y					

d) $y = \frac{1}{2}x + 1$

x	-4	-2	0	2	4
y					

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e) $y = 3x$

x	-2	-1	0	1	2
y					

f) $y = -\frac{1}{2}x + 3$

x	-4	-2	0	2	4
y					

C The equation from a table ●●○

Remember: The gradient is how much y changes for each step of 1 in x. Read c from the column where $x = 0$. Then write $y = mx + c$.

4. Each table shows points on a straight line. Write the equation of the line.

a)

x	-2	-1	0	1	2
y	-3	-1	1	3	5

Equation: _____

b)

x	-2	-1	0	1	2
y	-10	-7	-4	-1	2

Equation: _____

c)

x	-2	-1	0	1	2
y	5	4	3	2	1

Equation: _____

d)

x	0	1	2	3	4
y	5	3	1	-1	-3

Equation: _____

e)

x	-2	-1	0	1	2
y	-8	-4	0	4	8

Equation: _____

f)

x	0	1	2	3	4
y	-6	-5	-4	-3	-2

Equation: _____

D When the table does not show x = 0 ●●○

Remember: Find m first: the change in y divided by the change in x. Then put one point of the table into $y = mx + c$ and solve for c.

5. Write the equation of each line.

a)

x	2	3	4	5
y	7	9	11	13

Equation:

b)

x	3	4	5	6
y	8	11	14	17

Equation:

c)

x	1	2	3	4
y	7	5	3	1

Equation:

d)

x	1	3	5	7
y	1	5	9	13

Equation:

e)

x	2	4	6	8
y	4	2	0	-2

Equation:

f)

x	2	4	6	8
y	8	14	20	26

Equation:

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E Fractions, and tables that are not lines ●●●

> **Remember:** A gradient can be a fraction: if y goes up by 3 while x goes up by 2, the gradient is $\frac{3}{2}$. A table shows a straight line only when the y values change by the same amount every step.

6. Write the equation of each line. Give the gradient as a fraction.

a)

x	-2	-1	0	1	2
y	0	0.5	1	1.5	2

Equation:

b)

x	-2	0	2	4
y	-3	0	3	6

Equation:

c)

x	-4	-2	0	2	4
y	5	4	3	2	1

Equation:

d)

x	-3	0	3	6
y	-1	1	3	5

Equation:

7. Does each table show a straight line? Answer yes or no, with a reason.

a) Table A

x	0	1	2	3
y	1	3	5	7

b) Table B

x	0	1	2	3
y	1	4	9	16

c) Table C

x	0	1	2	3
y	5	4	3	2

F Tables in real situations ●●●

> **Remember:** The letters change but the idea is the same: the gradient is the change for each step, and the intercept is the starting value.

8. A taxi company charges a fixed amount plus a rate per kilometre. The table shows the fare F in dollars for a trip of d kilometres.

d (km)	0	1	2	3	4
F ($)	5	8	11	14	17

a) Write the equation that connects F and d.

b) Find F when $d = 7$.

c) Find d when $F = 35$.

d) What does the number 5 tell you in this situation?

e) What does the number 3 tell you in this situation?

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9. A candle is lit. The table shows its height h in centimetres after t hours.

t (hours)	0	1	2	3	4
h (cm)	20	18	16	14	12

a) Write the equation that connects h and t.

b) Find h when $t = 7$.

c) Find t when $h = 0$.

d) What does the number 20 tell you in this situation?

e) What does the number 2 tell you in this situation?

Extension

10. This table shows points on a straight line. Find its equation.

x	4	7	10	13
y	5	11	17	23

11. Two straight lines are shown by these tables. For which value of x do the two lines have the same y value? Find the equations first.

a) Line A

x	0	1	2	3
y	1	3	5	7

b) Line B

x	0	1	2	3
y	7	6	5	4

c) Same y when

12. The equation of a line is $y = 4x - 3$. Find the value of x that gives $y = 101$.

13. A line has gradient 3 and passes through the point $(2,\ 11)$. Find c, and write the equation of the line.

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Linear Graphs 1: Tables and the Equation

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KEY WORDS table of values *(tūtohi uara)* · gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c

A Reading a table ●○○

Remember: In a table for a straight line, y changes by the same amount each time x goes up by 1. That amount is the gradient m. The value of y when $x = 0$ is the y-intercept c.

1. The table shows some of the points on a straight line.

x	-2	-1	0	1	2	3
y	-9	-5	-1	3	7	11

a) What is y when $x = 2$? _____

b) Each time x goes up by 1, what happens to y? _____

c) What is y when $x = 0$? _____

d) Continue the pattern: what is y when $x = 4$? _____

e) Write the equation of the line. _____

2. The table shows some of the points on a straight line.

x	-1	0	1	2	3	4
y	11	8	5	2	-1	-4

a) What is y when $x = 2$? _____

b) Each time x goes up by 1, what happens to y? _____

c) What is y when $x = 0$? _____

d) Continue the pattern: what is y when $x = 5$? _____

e) Write the equation of the line. _____

B A table from the equation ●○○

Remember: Put each value of x into the equation to find y. Take care with negatives: $2 \times (-3) = -6$.

3. Complete the table of values for each line.

a) $y = 3x - 2$

x	-2	-1	0	1	2
y					

b) $y = -x + 5$

x	-2	-1	0	1	2
y					

c) $y = 6 - 2x$

x	-2	-1	0	1	2
y					

d) $y = \frac{1}{2}x - 1$

x	-4	-2	0	2	4
y					

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e) $y = -2x$

x	-2	-1	0	1	2
y					

f) $y = -\frac{1}{2}x + 2$

x	-4	-2	0	2	4
y					

C The equation from a table ●●○

Remember: The gradient is how much y changes for each step of 1 in x. Read c from the column where $x = 0$. Then write $y = mx + c$.

4. Each table shows points on a straight line. Write the equation of the line.

a)

x	-2	-1	0	1	2
y	-5	-2	1	4	7

Equation: _____

b)

x	-2	-1	0	1	2
y	-9	-7	-5	-3	-1

Equation: _____

c)

x	-2	-1	0	1	2
y	6	5	4	3	2

Equation: _____

d)

x	0	1	2	3	4
y	7	4	1	-2	-5

Equation: _____

e)

x	-2	-1	0	1	2
y	-10	-5	0	5	10

Equation: _____

f)

x	0	1	2	3	4
y	-8	-7	-6	-5	-4

Equation: _____

D When the table does not show x = 0 ●●○

Remember: Find m first: the change in y divided by the change in x. Then put one point of the table into $y = mx + c$ and solve for c.

5. Write the equation of each line.

a)

x	2	3	4	5
y	8	11	14	17

Equation:

b)

x	3	4	5	6
y	3	5	7	9

Equation:

c)

x	1	2	3	4
y	7	4	1	-2

Equation:

d)

x	1	3	5	7
y	1	7	13	19

Equation:

e)

x	2	4	6	8
y	7	3	-1	-5

Equation:

f)

x	2	4	6	8
y	5	9	13	17

Equation:

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E Fractions, and tables that are not lines ●●●

> **Remember:** A gradient can be a fraction: if y goes up by 3 while x goes up by 2, the gradient is $\frac{3}{2}$. A table shows a straight line only when the y values change by the same amount every step.

6. Write the equation of each line. Give the gradient as a fraction.

a)

x	-2	-1	0	1	2
y	-2	-1.5	-1	-0.5	0

Equation:

b)

x	-2	0	2	4
y	-2	1	4	7

Equation:

c)

x	-4	-2	0	2	4
y	4	3	2	1	0

Equation:

d)

x	-3	0	3	6
y	-3	-1	1	3

Equation:

7. Does each table show a straight line? Answer yes or no, with a reason.

a) Table A

x	0	1	2	3
y	1	2	4	8

b) Table B

x	0	1	2	3
y	3	5	7	9

c) Table C

x	0	1	2	3
y	10	7	4	1

F Tables in real situations ●●●

> **Remember:** The letters change but the idea is the same: the gradient is the change for each step, and the intercept is the starting value.

8. A plumber charges a call-out fee plus an hourly rate. The table shows the cost C in dollars of a job that takes h hours.

h **(hours)**	0	1	2	3	4
C **(\$)**	60	100	140	180	220

a) Write the equation that connects C and h.

b) Find C when $h = 6$.

c) Find h when $C = 380$.

d) What does the number 60 tell you in this situation?

e) What does the number 40 tell you in this situation?

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9. A tank is being drained. The table shows the volume of water V in litres after t minutes.

t (minutes)	0	1	2	3	4
V (litres)	500	475	450	425	400

a) Write the equation that connects V and t.

b) Find V when $t = 12$.

c) Find t when $V = 0$.

d) What does the number 500 tell you in this situation?

e) What does the number 25 tell you in this situation?

Extension

10. This table shows points on a straight line. Find its equation.

x	5	8	11	14
y	11	20	29	38

11. Two straight lines are shown by these tables. For which value of x do the two lines have the same y value? Find the equations first.

a) Line A

x	0	1	2	3
y	-2	1	4	7

b) Line B

x	0	1	2	3
y	10	9	8	7

c) Same y when

12. The equation of a line is $y = 5x + 2$. Find the value of x that gives $y = 102$.

13. A line has gradient 4 and passes through the point $(3,\ 14)$. Find c, and write the equation of the line.

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